

The NCLEX tests your ability to apply nursing knowledge, so it's important to understand **pathophysiology, assessment findings, nursing interventions, and patient education** for autoimmune diseases. Here's a breakdown:

1. GENERAL CONCEPTS OF AUTOIMMUNE DISEASES

- The immune system mistakenly attacks the body's own healthy cells.
- Causes are **multifactorial** (many different causes contribute), including genetics, infections, and environmental factors.
- Often **chronic** (long-term) and **progressive** (worsens over time).
- Treatment usually involves **immunosuppressants** (drugs that weaken the immune system) and **anti-inflammatory** medications.

2. COMMON AUTOIMMUNE DISEASES ON NCLEX

RHEUMATOID ARTHRITIS (RA)

- **Pathophysiology:** The immune system attacks **synovial joints** (joints with fluid that allows smooth movement), causing **inflammation** (swelling and redness) and **joint deformity** (permanent joint damage).
- **Symptoms:** Joint pain, swelling, morning stiffness lasting >1 hour, **nodules** (small lumps under the skin), systemic symptoms like fatigue and fever.
- **Treatment:**
 - **DMARDs** (Disease-Modifying Anti-Rheumatic Drugs) slow disease progression, e.g., **methotrexate**.
 - **NSAIDs** (Non-Steroidal Anti-Inflammatory Drugs) reduce pain and swelling.
 - **Corticosteroids** (steroids that lower inflammation) used short-term.

SYSTEMIC LUPUS ERYTHEMATOSUS (SLE)

- **Pathophysiology:** The immune system attacks multiple organs, including skin, joints, kidneys, and heart.
- **Symptoms:**
 - **Butterfly rash** (red rash on cheeks and nose).
 - Joint pain.

- **Nephritis** (kidney inflammation).
- **Pericarditis** (inflammation of the heart lining).
- Sensitivity to sunlight.
- **Treatment:**
 - **Hydroxychloroquine** (reduces lupus flares).
 - **Corticosteroids** for severe cases.
 - **Avoid sunlight**, use sunscreen.

MULTIPLE SCLEROSIS (MS)

- **Pathophysiology:** The immune system attacks the **myelin sheath** (protective covering of nerves), causing nerve damage.
- **Symptoms:**
 - **Muscle weakness**, difficulty walking.
 - **Diplopia** (double vision), **nystagmus** (rapid eye movements).
 - **Paresthesia** (numbness/tingling).
 - **Bladder dysfunction** (urinary urgency or retention).
- **Treatment:**
 - **Corticosteroids** for acute attacks.
 - **Immunomodulators** (medications that regulate immune response).
 - **Physical therapy** to maintain mobility.

MYASTHENIA GRAVIS (MG)

- **Pathophysiology:** The immune system attacks **acetylcholine receptors** (needed for muscle contractions), leading to **muscle weakness**.
- **Symptoms:**
 - **Ptosis** (drooping eyelid).
 - Weakness of face, throat, respiratory muscles.
 - **Myasthenic crisis** (severe muscle weakness → respiratory failure).
- **Treatment:**
 - **Pyridostigmine** (improves nerve-muscle communication).
 - **Corticosteroids** and **plasmapheresis** (removes harmful antibodies from blood).

GRAVES' DISEASE (HYPERTHYROIDISM)

- **Pathophysiology:** The immune system overstimulates the **thyroid gland** (controls metabolism), causing too much **thyroid hormone**.
- **Symptoms:**
 - **Exophthalmos** (bulging eyes).

- **Tachycardia** (fast heart rate), **hypertension** (high blood pressure).
- **Heat intolerance** (trouble in hot weather).
- Weight loss, **goiter** (enlarged thyroid).
- **Treatment:**
 - **Methimazole** or **PTU** (blocks thyroid hormone production).
 - **Beta-blockers** (slow heart rate).
 - **Radioactive iodine therapy** to shrink the thyroid.

HASHIMOTO'S THYROIDITIS (HYPOTHYROIDISM)

- **Pathophysiology:** The immune system destroys the thyroid, causing **low thyroid hormone**.
- **Symptoms:**
 - Fatigue, **bradycardia** (slow heart rate), cold intolerance.
 - Weight gain, **constipation** (slow bowel movements).
 - Dry skin, hair thinning.
- **Treatment:**
 - **Levothyroxine** (synthetic thyroid hormone replacement).

TYPE 1 DIABETES MELLITUS (T1DM)

- **Pathophysiology:** The immune system destroys **pancreatic beta cells** (cells that make insulin), leading to **insulin deficiency**.
- **Symptoms:**
 - **Polyuria** (excessive urination), **polydipsia** (excessive thirst), **polyphagia** (excessive hunger).
 - **Ketoacidosis** (high blood sugar with acid buildup).
- **Treatment:**
 - **Insulin therapy** (lifelong).
 - **Monitor blood sugar** regularly.

CELIAC DISEASE

- **Pathophysiology:** The immune system reacts to **gluten** (protein in wheat, barley, and rye), damaging the **small intestine**.
- **Symptoms:**
 - **Diarrhea, bloating, steatorrhea** (fatty stools).
 - **Malabsorption** (poor nutrient absorption).
 - **Weight loss, anemia** (low red blood cells).
- **Treatment:**
 - **Gluten-free diet** for life.

3. NCLEX PRIORITIES FOR AUTOIMMUNE DISEASES

- **Patient Education:**
 - Avoid **triggers** (stress, infections, sun exposure for lupus).
 - Importance of **medication compliance**.
- **Prevent Infections:** Many treatments **weaken immunity**, increasing **infection risk**. Teach hand hygiene, vaccinations (except live vaccines in immunosuppressed patients).
- **Manage Fatigue:** Encourage **rest periods** to prevent overexertion.
- **Monitor for Complications:**
 - **RA → Joint deformities.**
 - **Lupus → Kidney failure.**
 - **MS → Paralysis.**
 - **MG → Respiratory failure.**
 - **Graves' Disease → Thyroid storm** (life-threatening excess thyroid hormone).